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Sulfinylamines

Regio-, Stereo-, and Enantioselective Copper-Catalyzed Allylation of Sulfinylamines with Allenes

Jie Zhu, Yue Zhao, Shengming Ma, and Minyan Wang**

Abstract: Sulfinamides, pivotal scaffolds in modern organic synthesis, have undergone a continuous evolution of synthetic methodologies over the past century, yet the precise construction of sulfur stereocenters remains a significant synthetic challenge. Herein, we report a versatile catalytic asymmetric strategy for the synthesis of S-chirogenic allyl sulfinamides through the direct allylation of sulfinylamines with allenes. Our methodology employs a catalytic system comprising copper salts, hydrosilanes, and readily accessible chiral ligands, which facilitates the in situ generation of organocopper surrogates via hydrocupration of allenes. This approach demonstrates remarkable substrate scope and excellent regio-, stereo-, and enantioselectivity across a diverse array of sulfinamide products. Comprehensive mechanistic investigations, including detailed experimental studies and computational analyses, have been conducted to elucidate the reaction pathway and identify the key factors governing enantioselectivity.

Chiral sulfinamides featuring stereogenic S(IV) centers have garnered particular attention as pivotal synthetic targets.[1–15] These compounds not only function as crucial intermediates for the stereoselective assembly of sulfur-containing functional groups[16–19] but also exhibit remarkable versatility as bifunctional reagents under mild photoredox catalysis conditions.[20,21] Among these, the synthesis of chiral S-allyl sulfinamides has garnered significant attention due to their intricate structural complexity and diverse applications. A landmark achievement in 1990 facilitated the enantiopure synthesis of S-allyl sulfinamides via ene reactions of *trans*-alkenes with *N*-sulfinylcarbamates derived from chiral alcohols (Figure 1a).[22] To eliminate the reliance on chiral auxiliaries, Beyah and colleagues in 2017 pioneered an asymmetric catalytic variant, employing a chiral antimony catalyst in the ene-type sulfinamidation of *cis*- alkenes (Figure 1b).[23] However, the propensity of the *N*-sulfonyl robust sulfinamide moiety to act as a leaving group, has limited its utility in adaptable synthetic methodologies.

Sulfinylamines (R–NSO), distinguished by their orthogonal cumulative π-systems, have emerged as pivotal building blocks in organic synthesis, facilitating a diverse range of chemical transformations.[24,25] The nucleophilic addition of magnesium reagents to sulfinylamines, developed in 1974, offers a direct method for synthesizing *S*-allyl sulfinamides, as well as a broad range of other sulfinamides (Figure 1c).[26–28] Subsequent catalytic advancements have aimed at enhancing compatibility with sensitive functional groups by employing readily accessible alkyl[29,30] and aryl[31] feedstocks in addition reactions. The advent of asymmetric catalysis utilizing aryl boron compounds[32,33] or halides[34–37] with sulfinylamines has introduced novel pathways for the synthesis of chiral sulfinamides. Inspired by these developments, we propose an enantioselective nucleophilic addition of allylmetal species to sulfinylamines to establish an *S*-stereogenic center. Here, we present an asymmetric catalytic allylation of sulfinylamines for the synthesis of enantioenriched *S*-allyl sulfinamides with excellent regio-, stereo-, and enantioselectivity (Figure 1d). Our strategy leverages copper salts, hydrosilanes, and commercially available ligands to in situ generate allylcopper nucleophiles from allenes with high regio- and stereoselectivity through hydrocupration.[38–68] Mechanistic investigations reveal that the formation of the enantioenriched *S*-allyl sulfinamides is achieved through the migratory insertion of the allylcopper nucleophile into the sulfinimine, rather than via a chair-like Zimmerman-Traxler transition state.[69,70] This approach not only enhances the synthetic efficiency but also broadens the scope of accessible chiral sulfinamides, paving the way for further exploration in sulfur-centered stereochemistry. Our investigation commenced with the utilization of allene **1a** with *N*-sulfinyltritylamine (**I**) as the model substrates (Table 1). We initially established a reaction system employing 10 mol% of $Cu(OAc)_2$ and 12 mol%

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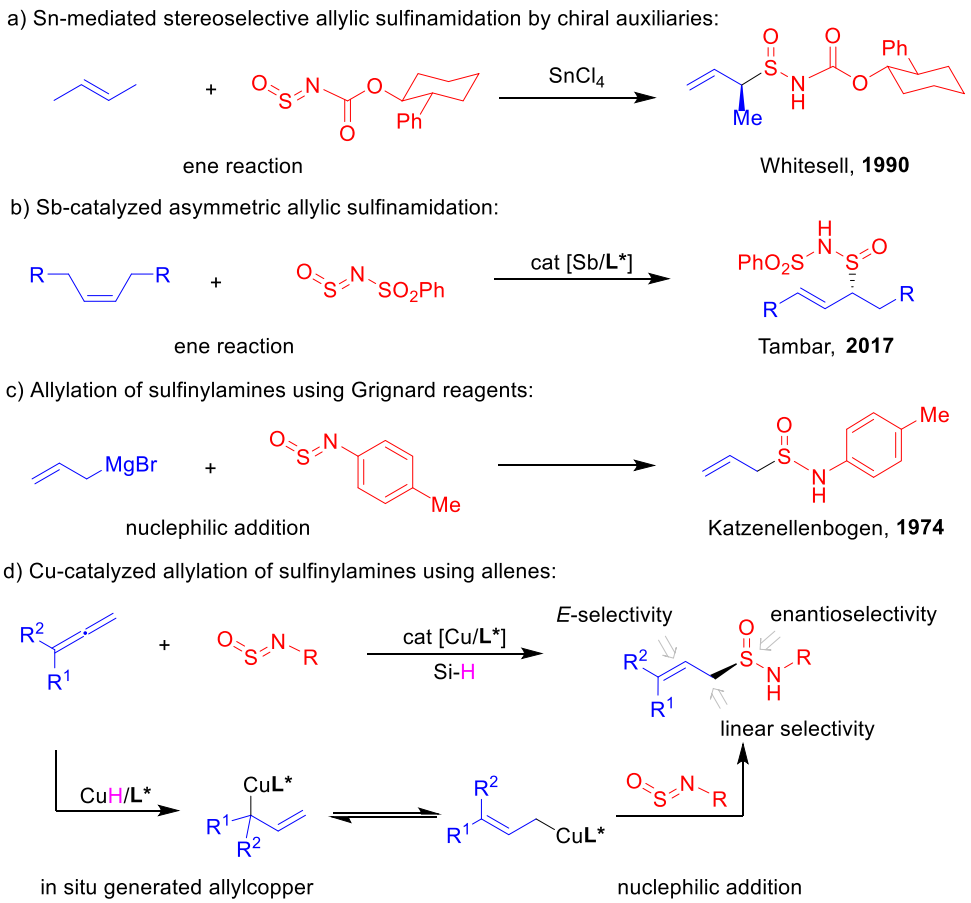


Figure 1. Development of asymmetric catalytic strategies for the construction of *S*-chirogenic allyl sulfinamides.

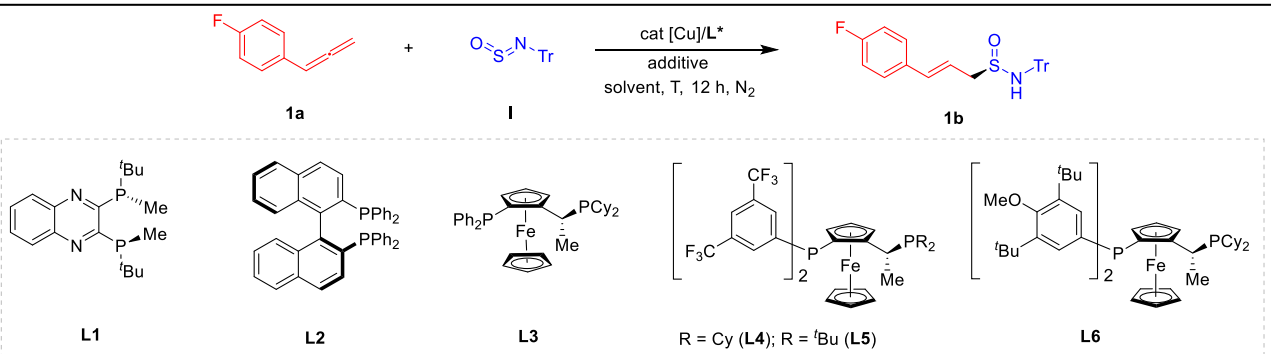
of QuinoxP (**L1**), along with 3.0 equivalents of DEMS, conducted in THF at 0 °C for 12 h under a nitrogen atmosphere. This preliminary setup afforded the target *trans* product **1b** in a modest 27% yield with an enantiomeric ratio (er) of 55/45 (entry 1). No rearrangement-type branched byproducts were observed. Subsequent ligand optimization revealed significant improvements. Switching to BINAP (**L2**) enhanced the yield to 37% with an improved er of 60/40 (entry 2). The introduction of JosiPhos (**L3**) further refined the er to 63/37 (entry 3). A more pronounced enhancement in enantioselectivity was observed upon substituting the phenyl groups on the phosphorus of JosiPhos with 3,5-CF_3-Ph groups (**L4**), which escalated the er to 85/15 while maintaining high yield (entry 4). However, replacing PCy_2 with P^tBu_2 (**L5**) resulted in a slight decline in both yield and enantioselectivity (entry 5). The use of ligand **L6**, featuring bulky electron-donating aryl groups on the phosphorus atom, led to a notable decrease in both yield and enantioselectivity (entry 6). The choice of solvent emerged as a critical factor for achieving high enantioselectivity. Employing dioxolane as the solvent yielded comparable enantioselectivity but reduced the yield to 30% (entry 7). Conversely, MTBE as the solvent not only improved the enantioselectivity to 87/13 er but also increased the yield to 87% (entry 8). Exploration of alternative copper sources, such as MesCu and TcCu, proved beneficial, with TcCu notably enhancing the yield to 89% with an er of 88/12 (entries 9 and 10). Various silane variants were also tested. DPS displayed relevant reactivity, while TES produced product **1b** with an er of 90/10 and a yield of 74% (entries 11 and 12). Further optimization revealed that lowering the reaction temperature enhanced the reaction outcome (entries 13 and 14). Remarkably, reducing the temperature to −60 °C resulted in an outstanding er of 95/5 and a yield of 80% (entry 15). Under these conditions, substituting TES with DEMS also yielded comparable results (entry 16). Reducing the catalyst loading to 5 mol% decreased conversion but preserved enantioselectivity (entry 17). Control experiments highlighted the indispensable role of the copper catalyst, as its omission completely halted the reaction (entry 18).

In our pursuit to comprehensively explore the reactivity and structural diversity of allenes, we conducted a systematic series of experiments utilizing sulfinylamine **I** as the key reactant partner (Table 2). Our investigation commenced with the evaluation of allene **2a** and its methyl-substituted analogues (**2a**–**7a**), followed by ethyl (**8a**), isopropyl (**9a**), *tert*-butyl (**10a**), and phenyl (**11a**) derivatives, all of which successfully yielded the corresponding products (**2b**–**11b**) with remarkable efficiency. Significantly, allenes bearing diverse ether functionalities on the benzene core (**12a**–**17a**) demonstrated exceptional reactivity, delivering the target products **12b**–**17b** with notably high enantioselectivities. The compatibility

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Table 1: Optimization of reaction conditions.[a)]



Entry	Cat [M] (mol%)	L* (mol%)	Additive (equiv.)	T (°C)	Solvent	er of **1b** (%)[b)]	Yield of **1b** (%)[c)]
1	$Cu(OAc)_2$ (10)	**L1** (12)	DEMS (3.0)	0	THF	55/45	27
2	$Cu(OAc)_2$ (10)	**L2** (12)	DEMS (3.0)	0	THF	60/40	37
3	$Cu(OAc)_2$ (10)	**L3** (12)	DEMS (3.0)	0	THF	63/37	71
4	$Cu(OAc)_2$ (10)	**L4** (12)	DEMS (3.0)	0	THF	85/15	86
5	$Cu(OAc)_2$ (10)	**L5** (12)	DEMS (3.0)	0	THF	83/17	73
6	$Cu(OAc)_2$ (10)	**L6** (12)	DEMS (3.0)	0	THF	61/39	41
7	$Cu(OAc)_2$ (10)	**L4** (12)	DEMS (3.0)	0	dioxolane	85/15	30
8	$Cu(OAc)_2$ (10)	**L4** (12)	DEMS (3.0)	0	MTBE	87/13	87
9	MesCu (10)	**L4** (12)	DEMS (3.0)	0	MTBE	86/14	84
10	TcCu (10)	**L4** (12)	DEMS (3.0)	0	MTBE	88/12	89
11	TcCu (10)	**L4** (12)	DPS (3.0)	0	MTBE	85/15	80
12	TcCu (10)	**L4** (12)	TES (3.0)	0	MTBE	90/10	74
13	TcCu (10)	**L4** (12)	TES (3.0)	−20	MTBE	90/10	82
14	TcCu (10)	**L4** (12)	TES (3.0)	−40	MTBE	93/7	82
15	**TcCu (10)**	**L4 (12)**	**TES** (3.0)	**−60**	**MTBE**	**95/5**	**80 (78)**[d)]
16	TcCu (10)	**L4** (12)	DEMS (3.0)	−60	MTBE	95/5	77
17	TcCu (5)	**L4** (6)	TES (3.0)	−60	MTBE	93/7	55
18	–	**L4** (12)	TES (3.0)	−60	MTBE	–	0

[a)] Reaction conditions: cat [Cu] (5–10 mol%), **L** (6–12 mol%), sulfinylamine **I** (0.1 mmol), **1a** (0.30 mmol), and hydrosilane (0.3 mmol) in 2 mL of solvent for 12 h under nitrogen atmosphere. [b)] Determined by chiral HPLC. [c)] Determined by crude ^{19}F NMR. [d)] Isolated yield. DEMS = $(EtO)_2MeSiH$. DPS = Ph_2SiH_2. TES = $(EtO)_3SiH$. MTBE = methyl *tert*-butyl ether. MesCu = (2,4,6-trimethylphenyl)copper. TcCu = copper(I) thiophene-2-carboxylate.

of halide-substituted substrates was particularly noteworthy, with fluorine (**18a–21a**), chlorine (**22a**), and bromine (**23a**) derivatives exhibiting minimal interference in the reaction outcomes while maintaining superior enantioselectivity. Our experimental results further confirmed the remarkable compatibility of fluoroalkyl substituents, including CF_3 (**26a**) and OCF_2H (**27a**), under the optimized reaction conditions. The treatment of naphthalene and dibenzo[*b*,*d*]thiophene-containing allenes (**26a–28a**) proceeded with commendable efficiency, yielding products (**26b–28b**) with excellent enantioselectivity. A particularly significant advancement was achieved in the successful transformation of 1,1-disubstituted allenes **29a–37a** featuring methyl and aryl moieties, which afforded products **29b–37b** with an exceptional enantioselectivity and *Z*/*E* selectivity. To unambiguously determine the stereochemical outcome, we performed X-ray crystallographic analysis on product **34b**,[71] which conclusively established its *E*,*R*-configuration. Expanding the substrate scope further, we demonstrated the exceptional chemoselectivity of the reaction towards naphthyl (**38a**), long-chain alkyl (**39a**), and cyclic (**40a**) allenes, which exclusively yielded the desired products **38b–40b** under the optimized conditions.

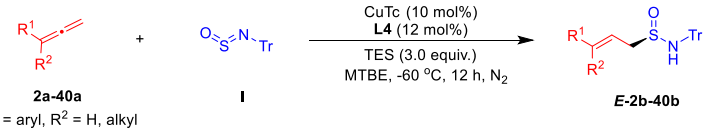
To demonstrate the synthetic versatility of our method, we conducted comprehensive experiments (Scheme 1). Initially, this approach's utility was highlighted through a 1.0 mmol-scale synthesis and subsequent transformations. The reaction using allene **1a** and sulfinylamine **I** as substrates was smoothly executed, yielding product **1b** in 68% yield without any loss of enantioselectivity. Subsequently, the deprotection experiment was performed. Treatment of sulfinamide **1b** with methanesulfonic acid led to the deprotection product **41** in 71% yield, maintaining the enantioselectivity. Hydrogenation of **1b** with a Pd/C catalyst produced *alkyl*-sulfinamide **42** in 80% yield with 94/6 er.

To gain deeper insights into the reaction pathway, we conducted extensive mechanistic studies as depicted in Scheme 2. Initially, when stoichiometric amounts of ligand **L4** and TcCu were combined, the copper complex **43** was formed, as confirmed by X-ray analysis[71] (Scheme 2a). Utilizing this complex as a catalyst in the reaction between substrates **1a** and **I** produced results of comparable efficacy, strongly suggesting that this complex serves as the active catalytic species. A clear linear relationship was observed when compounds **1a** and **I** were reacted using **L4** as the chiral

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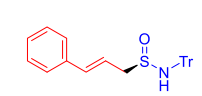
Table 2: Substrate scope of allenes.[a)]



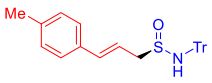
2a-40a
R^1 = aryl, R^2 = H, alkyl

I

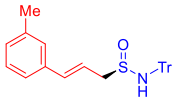
***E*-2b-40b**



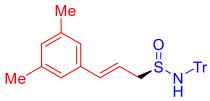
2b
65%, 95/5 er



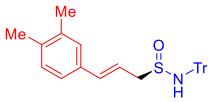
3b
79%, 95/5 er



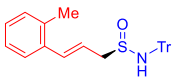
4b
66%, 95/5 er



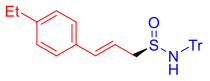
5b
65%, 95/5 er



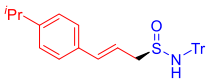
6b
80%, 95/5 er



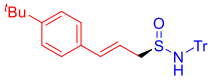
7b
71%, 95/5 er



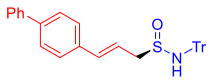
8b
65%, 95/5 er



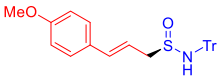
9b
54%, 95/5 er



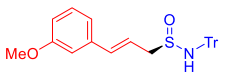
10b
81%, 94/6 er



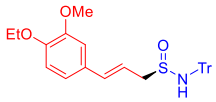
11b
80%, 93/7 er



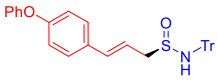
12b
72%, 96/4 er



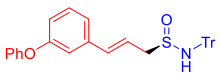
13b
45%, 95/5 er



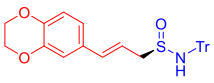
14b
47%, 97/3 er



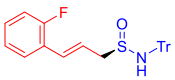
15b
80%, 93/7 er



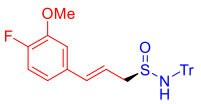
16b
58%, 93/7 er



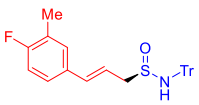
17b
45%, 93/7 er



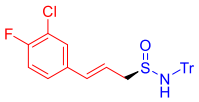
18b
73%, 91/9 er



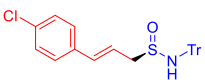
19b
72%, 95/5 er



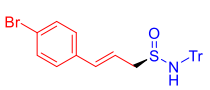
20b
93%, 92/8 er



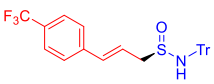
21b
72%, 92/8 er



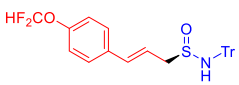
22b
70%, 91/9 er



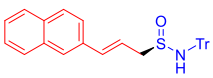
23b
70%, 95/5 er



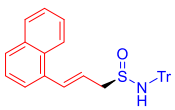
24b
39%, 95/5 er



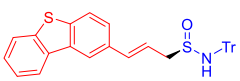
25b
74%, 95/5 er



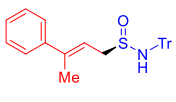
26b
59%, 94/6 er



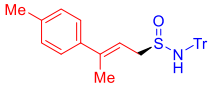
27b
51%, 92/8 er



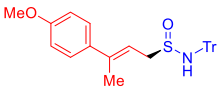
28b
73%, 93/7 er



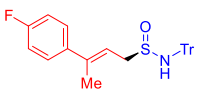
29b
55%, 97/3 er



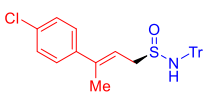
30b
57%, 95/5 er



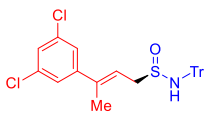
31b
47%, 93/7 er



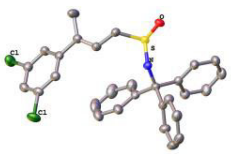
32b
60%, 94/6 er



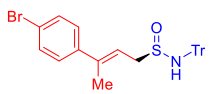
33b
55%, 96/4 er



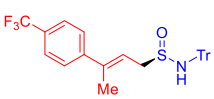
34b
45%, 99/1



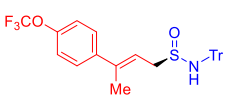
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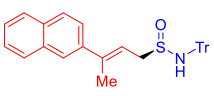
35b
56%, 98/2 er



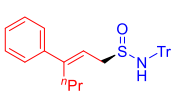
36b
51%, 99/1



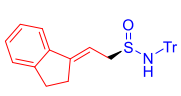
37b
68%, 96/4 er



38b
50%, 95/5 er



39b
42%, 92/8 er

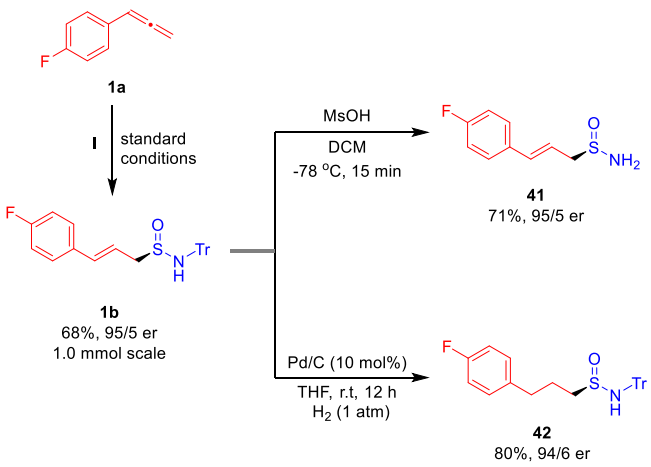


40b
28%, 95/5 er

[a)] Reaction conditions: TcCu (10 mol%), **L4** (12 mol%), sulfinylamine **I** (0.10 mmol), **2–40a** (0.30 mmol), TES (0.30 mmol), in MTBE, at −60 °C, 12 h, under N_2 atomsphere, in a sealed Schlenk tube. er values determined by chiral HPLC. Isolated yields determined by silica gel chromatography.

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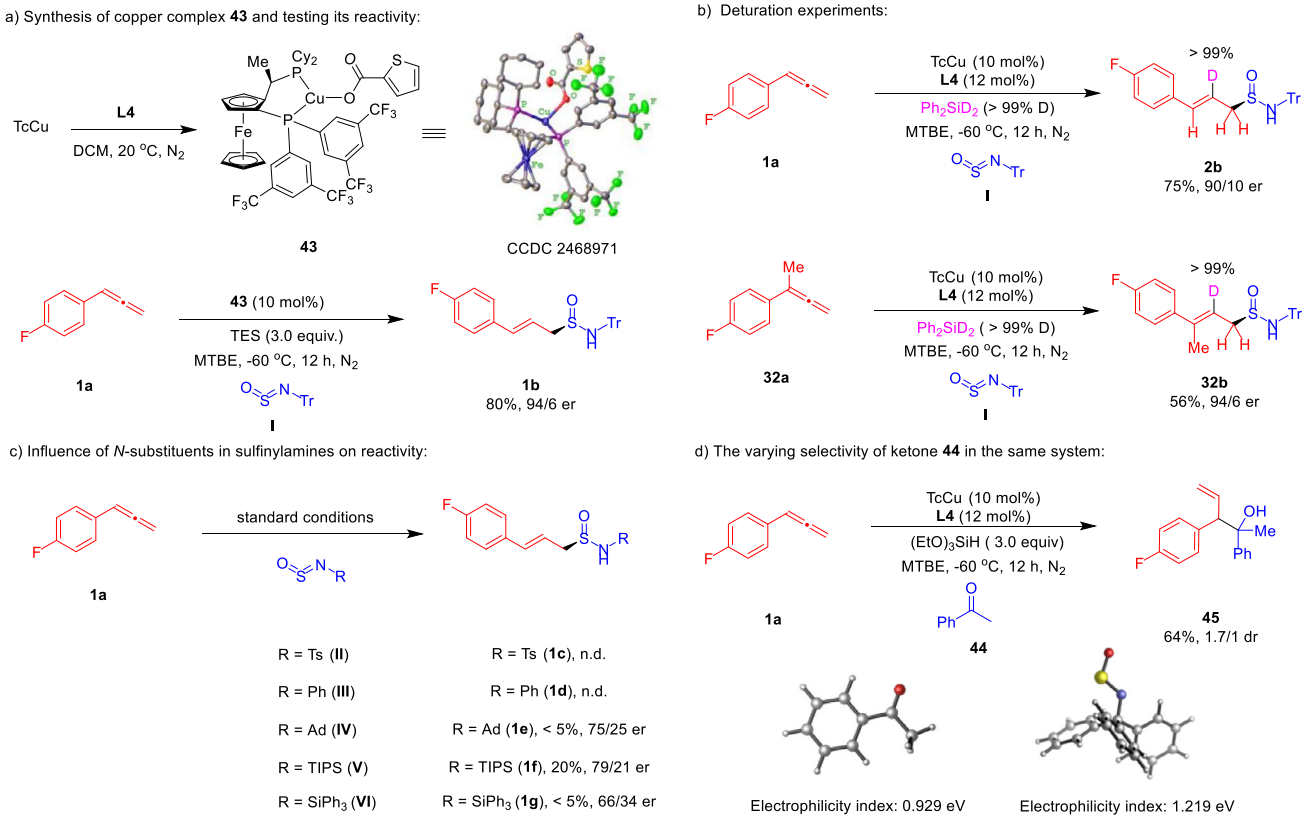
Scheme 1. Downstream transformations of **1b**.

ligand, indicating that the ratio of Cu to **L4** is 1/1, without any self-aggregation of the catalyst (see Supporting Information for details).

To further investigate the mechanism, we conducted deuterium labeling experiments (Scheme 2b). When Ph_2SiD_2 reagent was used instead of $(EtO)_3SiH$ with allene **1a** or **34a,** the D atoms were completely transferred to the central carbon of the allene, suggesting that the Cu-D species inserts at this position. Finally, the influence of different *N*-substituents on the sulfinylamines was examined (Scheme 2c).

Sulfinylamines with Ts (**II**) and Ph (**III**) substituents did not yield the desired products **1c** and **1d**. Although the use of sulfinylamines bearing sterically hindered Ad (**IV**), TIPS (**V**), and $SiPh_3$ (**VI**) groups produced small amounts of products **1e**–**g**, the enantioselectivity was low. These findings suggest that the choice of sulfinylamine **I** is advantageous for achieving both high reactivity and stereoselectivity. Notably, allene **1a** undergoes allylation with ketone **44** under the current reaction conditions, exclusively yielding the branched product **45** (Scheme 2d).[72–79] The notable regioselectivity difference underscores the distinct reactivity between ketones and sulfinylamines. Ketone **44** has an electrophilicity index of 0.929 eV, much lower than sulfinylamine's 1.219 eV. This higher index in sulfinylamine **I** reflects its stronger electrophilic nature, boosting its ability to attract electrons in reactions. Furthermore, the frontier molecular orbital theory is used to explain the different selectivities of the reaction (see Supporting Information for details).

Based on the aforementioned mechanistic experiments, density functional theory (DFT) was then performed to elucidate the detailed mechanism and the origin of stereo- and enantio-selectivity in the Cu(I)-catalyzed reaction between allene and sulfinylamine **I**.[80,81] The calculated reaction mechanism for the generation of allylcopper intermediates is illustrated in Supporting Information. Initially, the copper(I) catalyst coordinates with the chiral ligand **L4**, generating the active catalyst **43**. When silane is introduced into the system, the active catalyst **43** undergoes a σ-CAM mechanism with the silane via the transition state **TS1A**, with an activation



Scheme 2. Mechanistic experiments.

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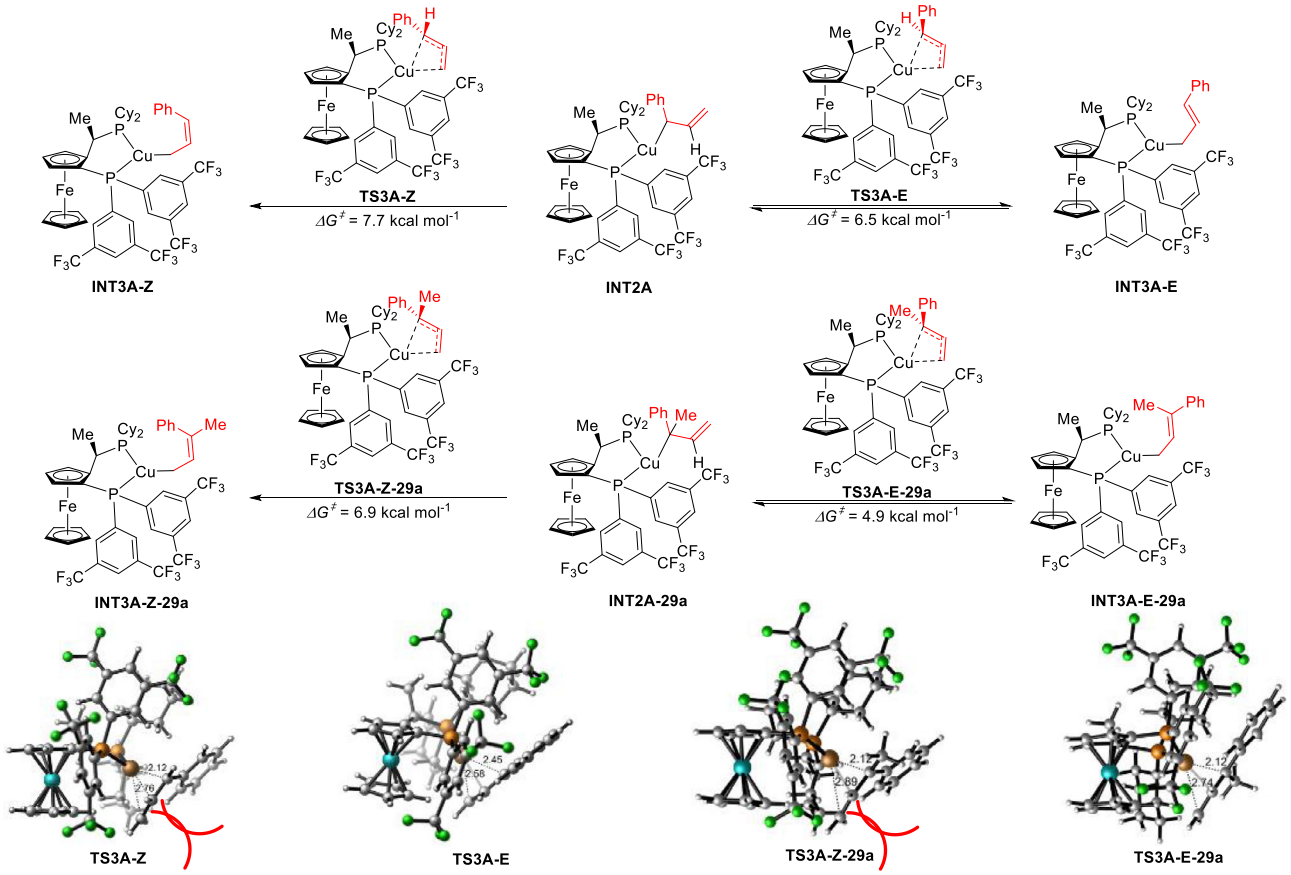


Figure 2. Seteroselectivity analysis for 1,3-copper migration.

energy of 18.5 kcal mol^{-1}.[82] The resulting copper-hydride species, **INT1A**, selectively engages in migratory insertion with the allene, leading to the formation of diverse types of organocopper species with regioselectivity (see Supporting Information for details).[83] Subsequently, the favorable intermediate **INT2A** undergoes a 1,3-copper migration, a key step in determining the stereoselectivity of the product (Figure 2).[84] For allene **1a**, the energy barrier for the formation of the *E*-configured intermediate **INT3A-E** via the transition state **TS3A-E** is 6.5 kcal mol^{-1}, which is 1.2 kcal mol^{-1} lower than that for the formation of **INT3A-Z** via transition state **TS3A-Z** (6.5 versus 7.7 kcal mol^{-1}). For the disubstituted allene **29a**, the stereoselectivity of the reaction can also be well controlled (4.9 kcal mol^{-1} for **TS3A-E-29a** versus 6.9 kcal mol^{-1} for **TS3A-Z-29a**). These disparities are primarily attributed to the substantial steric hindrance between the phenyl group and the alkenyl hydrogen in both **TS3A-Z and TS3A-Z-29a**.

Furthermore, we investigated the allylation reactions of **INT2A** and **INT3A-E** with sulfinylamine **I**, respectively (Figure 3). The coordination of N and O atoms of sulfinylamine into intermediate **INT2A** can form **INT3A-N** and **INT3A-O**, respectively. Due to the relatively higher calculated energy for **INT3B-N**, we excluded the reaction pathway involving this intermediate. The allylation reaction that proceeds through the six-membered chair-like Zimmerman-Traxler transition state **TS3B-linear-R** has an energy barrier of 11.8 kcal mol^{-1}. Although this pathway is more kinetically favorable than the direct allylation route leading to branched products, its energy requirement remains significantly higher than that of the 1,3-copper migration process of **INT2A** via transition states **TS3A-E**. These computational results imply that **INT2A** exhibits a distinct preference for undergoing transformation to yield **INT3A-E**, rather than participating in an allylation reaction with sulfinylamine. The migratory insertion between intermediate **INT3A-E** and sulfinylamine **I** is the crucial step dictating both the regio- and enantioselectivity within the catalytic cycle. The migratory insertion reaction of the S=N bond through the transition state **TS4A-linear-R** manifests a pronounced kinetic favorability. Specifically, the energy barrier associated with this pathway is 2.8 kcal mol^{-1} lower than that of the analogous reaction pathway via the transition state **TS4A-linear-S** (4.0 kcal mol^{-1} versus 6.8 kcal mol^{-1}). This distinct energy disparity leads to a greater tendency for the product to form in the *R*-configuration. The selective migratory insertion of the S=O double bond into intermediate **INT3A-E** occurs via the transition states **TS4B-R** and **TS4B-S**, with activation energies of 8.2 and 5.4 kcal/mol, respectively. These values are substantially higher than the activation energies associated with the migratory insertion involving the S=N double bond. To gain deeper insights into the enantioselectivity observed in the migratory insertion process, the independent gradient model based on Hirshfeld partitioning (IGMH) was utilized to

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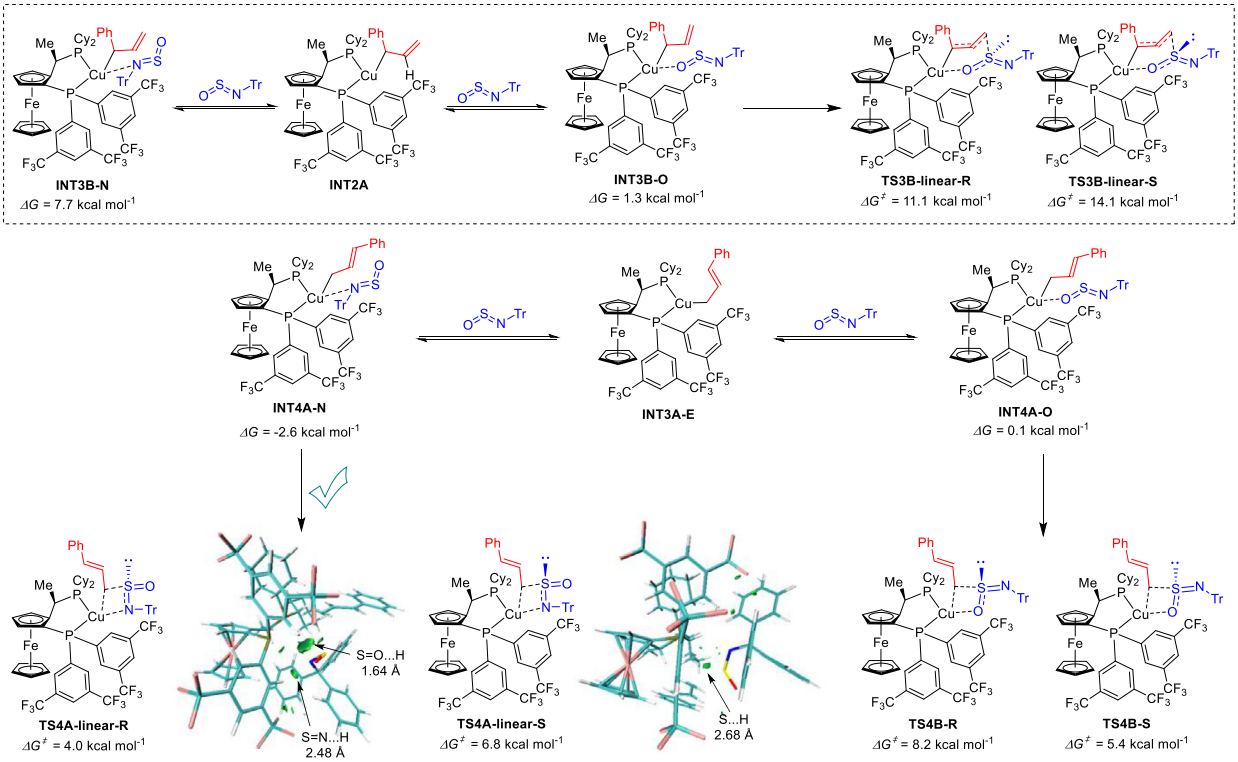


Figure 3. Transition-state models of regiodivergence in copper-catalyzed allylation reactions.

analyze the non-covalent interactions between the substrate and the chiral ligand fragment in transition states **TS4A-linear-R** and **TS4A-linear-S**.[85] Structural analysis reveals that in the favorable transition state **TS4A-linear-R**, stronger hydrogen-bond interactions exist between the C-H bonds of the chiral ligand and the oxygen or nitrogen atoms of reagent **I**. These interactions contribute to the stabilization of the transition state **TS4A-linear-R**, resulting in an activation energy barrier 2.8 kcal mol^{-1} lower than that of the transition state **TS4A-linear-S**.

In conclusion, this study establishes a robust and versatile copper-catalyzed platform for the stereoselective synthesis of chiral *S*-allyl sulfinamides, addressing a long-standing challenge in asymmetric sulfur chemistry. The developed methodology demonstrates remarkable functional group tolerance and substrate scope, enabling the efficient coupling of diverse allenes with sulfinylamines under mild reductive conditions. Through a synergistic combination of experimental investigations and computational analyses, we have elucidated the intricate reaction mechanism and identified key factors governing the selective aryl addition processes. Given the growing recognition of chiral *S*-stereogenic compounds as privileged structural motifs in pharmaceutical development and agrochemical innovation, we anticipate that this catalytic system will find broad applications in the synthesis of biologically active molecules.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: Allenes · Allylation · Asymmetric catalysis · Copper · Sulfinamides

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